The universe as a thing: Brading's P1** and the limits of local presentism

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Abstract

Brading's local present is a spatiotemporal region of whatever size is needed to sustain a dynamical system. Her P1** requires every existing thing's present to be less than the whole history, and she says that "in order to avoid the block universe, all we need is (a)." This note argues that her own principle makes the argument fail for the universe itself. Brading holds that what exists are the genuine unities, and that the law-constitutive principle applies "all the way up to the material universe as a whole." The universe is therefore a genuine unity and a thing that exists; its sustaining region is the whole history; so clause (a) is false of it. The alternatives—a spatially global present, a local or point present, or the universe's non-existence—are each excluded by Brading's text. The 2013 defeat condition does not cover the case that matters: the universe can have the whole history as its region while genuine subsystems have proper regions. The empirical question about subsystems is therefore not the question on which the view turns. A second argument shows that even for proper subsystems, (a) and (b) do not imply a proper union; a two-system, four-event model satisfies both and covers the history. The missing premise is not another bound on the extent of presents but a selection of which unities count—a selection Brading's scale-free principle cannot make. The verdict is limited: P1** is either false of the universe or restricted in a way that gives up the global presentist claim.

1 The texts and the tension

Katherine Brading's *Presentism as an Empirical Hypothesis* and *Physically Locating the Present* offer one of the most careful attempts to keep a presentist ontology alive within special relativity (Brading, 2013, 2015). She rejects the assumption that spacetime is the ontological principle of unity of what exists, and replaces it with a law-constitutive principle: the dynamical laws constitute genuine unities, and the present of a system is the spatiotemporal region needed to sustain its dynamics. The 2013 paper introduces the idea in a compact form:

> The present is a spatiotemporal region of whatever size is necessary to sustain the dynamical system in question (Brading, 2013, p. 1109).

The size is left open. It "might turn out to be much, much smaller than the abstract spatiotemporal structure within which the evolution of that unity is completely described, or it might be the size of the block universe" (Brading, 2013, p. 1109). Brading then assumes genuine subsystems and draws the local conclusion:

> So let us begin from the position that there are genuine subsystems. This means that the size of the spatiotemporal region required to sustain the system is less than the

entire block universe, and so the present is (at least in the first instance) local, not global (Brading, 2013, p. 1110).

The 2015 paper states the position as P1**:

(P1**) For each and every thing that exists, that thing exists only presently, where for each thing the spatiotemporal extent of the present for that thing is dependent on the dynamics and is (a) less than the entire spatiotemporal evolution of the universe as represented in Minkowski spacetime, and (b) less than the entire spatiotemporal evolution of that thing as represented in Minkowski spacetime (Brading, 2015, p. 17).

She adds: "In order to avoid the block universe, all we need is (a)" (Brading, 2015, p. 17). Her own statement of the effect is weaker: a smaller-than-universe present "calls the ontology of a block universe into question" (Brading, 2015, p. 17). And she leaves the empirical question open: whether there are genuine subsystems "smaller in spatiotemporal extent than the entirety of spacetime" is a matter of the details of the dynamics, and "only if there are such can the version of presentism offered here survive" (Brading, 2015, p. 18).

The tension is internal. On the same pages Brading commits herself to a principle that makes the material universe a genuine unity and hence a thing that exists. P1** then applies to the universe itself. But the region needed to sustain the universe is the whole history, so clause (a) is false of it. The rest of this note states the argument, answers the obvious replies, and draws the limited consequence.

The note does not claim that the coverage problem is wholly new, and it does not compete with Savitt's 2021 objection. It claims that the totality argument is distinct from Savitt's, that it is generated by Brading's own texts, and that P1** cannot be true of the universe as a whole. The formal model in §5 is a second, independent problem for the subsystem family.

2 What exists are the genuine unities

Brading's law-constitutive account is not only an account of the unity of ordinary objects. She says that the principle applies across the board:

We are being offered a principle of ontological unity that applies across the board to simple things, composite systems, and component parts of those systems, and which applies all the way up to the material universe as a whole. Insofar as what exists is what is one, the law-constitutive approach to unity is a metaphysical thesis about what exists (Brading, 2015, p. 16).

The "insofar as" might look like a hedge, but the surrounding text removes the doubt. Brading says that she begins "from the premise that what exists (what is real) are the genuine unities," and that "what is 'present' depends on what is real (i.e. what exists, the genuine unities), and what is real depends on the dynamical laws" (Brading, 2015, p. 17). This is not a conditional aside; it is the order of priority that distinguishes her view from the traditional presentism she rejects. She first determines what is real via the law-constitutive approach, and then seeks features of what is real as the ground of what is present.

The universe is the paradigm case of the principle's scope. Brading says the principle applies "all the way up to the material universe as a whole" (Brading, 2015, p. 16), and her 2013 note says the law-constitutive approach covers "the unity of simple things, composite things, and the totality

of existing things" (Brading, 2013, p. 1106, n. 3). The universe is not a collection of genuine unities that falls outside the principle; it is the maximal case to which the principle is meant to apply.

Two consequences follow. First, the universe is a genuine unity. Second, by Brading's identification of what exists with the genuine unities, the universe is a thing that exists. It is not merely a set or a collection; it is the maximal unified thing. This is the premise that the rest of the argument uses.

The identification is also what makes the view a metaphysical thesis rather than a taxonomy of dynamical systems. If existence were not unity, the law-constitutive approach would tell us which systems are unified but not which of them are real. Brading's claim is stronger: the unified systems just are what there is. A reader who rejects the identification is not yet engaged with her view; a reader who accepts it must follow it to the universe.

3 The reductio

The argument has four premises, each sourced to Brading's text.

(T1) What exists are the genuine unities, and the law-constitutive principle applies to the material universe as a whole (Brading, 2015, p. 16).

(T2) The present of a thing is the spatiotemporal region necessary to sustain it (Brading, 2013, pp. 1109–1110); (Brading, 2015, p. 17).

(T3) P1** requires, for every existing thing, that this region be less than the entire spatiotemporal evolution of the universe (Brading, 2015, p. 17).

(T4) The region necessary to sustain the universe as a whole is the entire history. The universe's unity is constituted by the relations among its members (Brading, 2015, p. 16), and its members are spread through the history; a region that omitted any of them would omit part of what the universe is.

From (T1) and (T2), the universe is a genuine unity and has a present: the region necessary to sustain it. From (T3), that region must be less than the whole history $\mathcal{W}$. From (T4), it is $\mathcal{W}$. Contradiction. Therefore P1** (a) is false of the universe, and P1**, which quantifies over every thing that exists, is false.

The argument can be put as a dilemma about the universe's present. It is either $\mathcal{W}$, or a proper region. If it is $\mathcal{W}$, (a) is false. If it is proper, it is either spatially global or local. A spatially global present is rejected by Brading: "the idea of a spatially global present common to all systems is not preserved in this version of presentism, but the rejection of a spatially global present is one of the central lessons of special relativity" (Brading, 2015, p. 17). A local or point-like present is the position she associates with Stein's "peculiarly extreme (but pluralistic!) form of solipsism" (Brading, 2013, p. 1102), and she says her view avoids it (Brading, 2013, pp. 1109–1110). A proper region that is neither global nor local would have to sustain the universe's unity while omitting some of the relations among its members; that is not a region sufficient to sustain the unity, on Brading's own account of aggregation: "The members must stand in relations to one another, and those relations must be necessary and sufficient for the collection to form a unity" (Brading, 2015, p. 16).

The remaining escape is to deny that the universe is a thing that exists. That contradicts (T1). It also contradicts the order of priority Brading defends: what is real are the genuine unities, and

the universe is the maximal one. If the universe is not a thing, the principle does not apply to the case that most obviously satisfies it.

The conclusion is not that Brading's physics is wrong, or that presentism is false. It is that P1** is not a condition that the universe can satisfy. The one thing whose existence is entailed by the law-constitutive principle is the one thing for which clause (a) fails. "All we need is (a)" cannot be satisfied by the universe.

4 The defeat condition is incomplete

Brading's 2013 defeat condition makes the same point in a different form. She writes:

> It might turn out that, for whatever system we try to consider, the size of the spatiotemporal region necessary to sustain it is the entire history of the universe, encompassing all that ever has been and ever will be. In other words, the only dynamical system that there is, is this entirety, and there are no genuine subsystems of the universe. If that turns out to be the case, then this version of presentism is defeated and the block universe triumphs (Brading, 2013, p. 1109).

The passage addresses one scenario: every system we consider has the whole history as its sustaining region. In that scenario there are no proper subsystems. But it is not the only scenario in which the universe's region is $\mathcal{W}$. The universe can have $\mathcal{W}$ as its region while its subsystems have proper regions. The subsystems are parts of the universe, not alternatives to it. Brading's principle makes the universe a genuine unity; its region is $\mathcal{W}$; the subsystems can be genuine too. The defeat condition does not cover this case.

This is not a verbal slip. It is the reason the 2015 survival condition is incomplete. Brading says that "only if there are such [genuine subsystems smaller than the entirety of spacetime] can the version of presentism offered here survive" (Brading, 2015, p. 18). But the existence of such subsystems does not make P1** true of the universe. The universe remains a genuine unity with the whole history as its present. The survival condition is necessary for the local presents of subsystems; it is not sufficient for the truth of P1**, because P1** also applies to the universe.

The empirical question is therefore not the question on which the view turns. Brading frames the issue as whether there are genuine subsystems, and treats the answer as a matter for the details of physics. The totality argument says that the answer does not decide the fate of P1**: the universe is a counterexample whether or not subsystems exist. The empirical question may decide whether there are local presents for subsystems; it does not decide whether the universal clause is true.

5 The retreat to subsystems

The natural reply is to restrict P1** to proper subsystems. The universe is not a "thing" in the intended sense; the quantifier ranges over the genuine unities below the whole. This reply has two costs.

First, it gives up the global claim. P1** says "for each and every thing that exists." If the universe is excluded, the quantifier is restricted, and the view no longer says that every existing thing exists only presently. It says that every proper subsystem does. The universe itself is left out, and with it the presentist claim that the whole of reality is present. A restricted quantifier is a different thesis, not a reading of the original.

Second, even for proper subsystems, (a) and (b) do not imply a proper union. Clause (a) says, for every system i,

$$\mathcal{U}_i \subsetneq \mathcal{W}.$$

It does not imply

$$\bigcup_i \mathcal{U}_i \subsetneq \mathcal{W}.$$

Proposition 1 (Non-implication)**.** *Local boundedness of every $\mathcal{U}_i$ is consistent with $\bigcup_i \mathcal{U}_i = \mathcal{W}$.*

Proof. Take $\mathcal{W} = \{a, b, c\}$ and the proper subsets $\{a, b\}$ and $\{b, c\}$. Their union is $\mathcal{W}$. □

The natural reply is that clause (b) closes the gap. Clause (b) says that each thing's present is less than the entire spatiotemporal evolution of that thing. That is a bound on the present relative to the thing's own history. It is not a bound on the present relative to every other thing's present, and it is not a claim that some event lies outside every present.

Proposition 2 ((a) and (b) do not imply a proper union)**.** *There is a model in which (a) and (b) hold for every system and $\bigcup_i \mathcal{U}_i = \mathcal{W}$.*

Proof. Let $\mathcal{W} = \{a, b, c, d\}$. Let system 1 have history $\mathcal{H}_1 = \{a, b, c\}$ and present $\mathcal{U}_1 = \{a, b\}$. Let system 2 have history $\mathcal{H}_2 = \{b, c, d\}$ and present $\mathcal{U}_2 = \{c, d\}$. Then $\mathcal{U}_1 \subsetneq \mathcal{W}$ and $\mathcal{U}_2 \subsetneq \mathcal{W}$, so (a) holds. Also $\mathcal{U}_1 \subsetneq \mathcal{H}_1$ and $\mathcal{U}_2 \subsetneq \mathcal{H}_2$, so (b) holds. But $\mathcal{U}_1 \cup \mathcal{U}_2 = \{a, b, c, d\} = \mathcal{W}$. □

The model is set-theoretic, not dynamical. That is sufficient for the non-implication: if (a) and (b) are offered as the conditions that exclude the block, a model that satisfies them and covers the history shows that they do not, by themselves, exclude it. A defender who says that the model is not law-constituted must supply the further dynamical condition and show that it excludes coverage. That is precisely the missing premise, and it is not (a) or (b).

The failure is structural. Clause (b) puts events outside their *own* system's present. It does not put any event outside *every* present. A defender needs the latter: an event $e \in \mathcal{W}$ such that for all i, $e \notin \mathcal{U}_i$. Neither (a) nor (b) delivers such an event. Per-item properness is coverage-neutral. Non-coverage requires a joint constraint on the family, or a selection principle that removes some presents from the family. Brading supplies neither.

The two problems are independent. The totality argument says that P1** is false of the universe; the coverage argument says that, even if the universe is set aside, the subsystem clauses do not secure a proper union. A defender who restricts the quantifier to avoid the first falls into the second. A defender who adds a selection principle to avoid the second still has not answered the first, unless the principle also excludes the universe—which is the quantifier restriction again.

6 Why the principle cannot select a scale

The diagnosis is not that Brading lacks an existence criterion. She has one: what exists are the genuine unities (Brading, 2015, p. 17). The problem is that the criterion is scale-free. It applies to simple things, composite systems, component parts, and the material universe as a whole. A criterion that confers existence on every level of unity cannot, by itself, select a local level as the level of existence-conferring presents. It confers existence on the maximal unity too.

This is the sense in which the missing premise is a principle of *selection*, not a further bound on the extent of presents. To get the local presentist result, one needs to say which unities' presents

count. The options are familiar: a privileged actual system, a co-presence relation, or a frame- or level-relative existence predicate. Each is a further structure, and each has a cost.

A privileged actual system is a metaphysical posit. It is not derived from the dynamics, and if the privileged system is the universe, its present is $\mathcal{W}$ and the view yields the block. A co-presence relation answers Savitt's objection about shared presents, but it does not answer coverage: a relation can relate all the local presents while their union is still $\mathcal{W}$. Frame- or level-relative existence blocks coverage by changing the question; every event exists for some system or level, and the global presentist claim is given up. Balaguer's R-presentism and Mason's domain- and level-dependent empirical presentism develop versions of this route (Balaguer, 2026; Mason, 2026), and Rovelli's "neither presentism nor eternalism" defends a local, unoriented becoming (Rovelli, 2019). These are coherent positions, but they are not P1**.

The scale-free criterion also explains why the 2013 defeat condition is incomplete. Brading asks whether every system's region is the whole history, and concludes that there are no proper subsystems. But the criterion applies at every scale, so the maximal unity and its proper parts can both be genuine. The region of the whole can be $\mathcal{W}$ while the regions of the parts are proper. The defeat condition does not consider this case.

7 Relation to Savitt, Saunders and Pooley

Savitt's 2021 objection is distinct and remains unanswered. He argues that Brading's local present is "a characterization of a relativistic present, not relativistic presentism," because every dynamical system has a sustaining region and "it is unclear under what conditions two dynamical systems share a present" (Savitt, 2021, §3.3). That is a population and co-presence objection: many presents, no public present. It is not the totality argument. Savitt does not discuss Brading 2015, P1**, or the law-constitutive principle's application to the universe. The two arguments are compatible: Savitt gives the semantic verdict, and the totality argument gives the metaphysical reason internal to Brading's system.

Saunders's requirement that the presentist provide an invariant, definable equivalence relation of co-presence is the sharpest version of the co-presence cost (Saunders, 2000); the definability results make a non-trivial symmetric equivalence relation of a present hard to come by (Clifton and Hogarth, 1995). It supports Savitt's objection, not the totality argument. Pooley's distinction between a deflationary analysis of passage compatible with the block and a privileged global present that does not take relativity seriously (Pooley, 2013) is about passage, not about the domain of existence. Brading's paper is about existence, so Pooley's dilemma does not apply directly. The totality argument does not claim that local presents fail to account for passage.

The interaction passage is sometimes read as a system-relative existence claim. Brading writes: "we belong to the same world as whatever we interact with, and the rest of that world is as real as we are," and asks, "Does my son exist relative to me when he is in London and I am here? Of course" (Brading, 2013, p. 1110). The passage is better read as a single-domain claim: "the rest of that world is as real as we are" is not indexed to a system, and the following question is about the *determinacy* of what the son is doing, not about his existence. It does not supply a relativized quantifier, and it does not answer the totality argument.

8 Objections and limits

The "insofar as" hedge. A reader might say that Brading's "insofar as what exists is what is one" is conditional, and that she could withdraw the identification. The surrounding text makes this difficult. She says she begins "from the premise that what exists (what is real) are the genuine unities," and that "what is 'present' depends on what is real (i.e. what exists, the genuine unities)" (Brading, 2015, p. 17). The identification is the order of priority that distinguishes her view from the traditional presentism she rejects. If she withdraws it, the law-constitutive approach no longer tells us what exists; it tells us only which systems are unified. That is a different, weaker view, and it is not the one P1** states.

A proper local sustaining region. The weakest premise is (T4). A defender might say that the universe's unity is sustained by a proper region that is neither a global slice nor a point: a local patch whose laws determine the rest. If such a region exists, (a) could be true of the universe. The reply is that the region must contain the relations among the universe's members, and those members are spread through the history. A local patch that omits them omits part of what the universe is. The defender needs a principled account of how a proper, local region sustains the whole; the note does not claim that no such account can be given. It claims that Brading does not give one, and that her own rejection of the global slice and the point leaves no obvious candidate.

Priority. The note does not claim that the totality argument is wholly new, and it does not claim that no one has noticed the tension. It claims that the argument is generated by Brading's texts, that it is distinct from Savitt's, and that it has not been answered. A published statement of the same inconsistency would defeat the priority claim, not the argument.

What the note does not show. It does not show that presentism is false, that eternalism is true, or that Brading's physics is wrong. It does not show that there are no genuine subsystems. It shows that P1** is either false of the universe or restricted in a way that gives up the global claim, and that the subsystem clauses do not secure a proper union.

9 Conclusion

Brading's P1** is her strongest anti-block formulation. It is also, by her own principle, false of the universe. What exists are the genuine unities; the principle applies all the way up to the material universe; the universe is therefore a thing that exists. Its sustaining region is the whole history. Clause (a) requires that region to be less than the whole. The contradiction is not a defect of the physics; it is a defect of the formulation.

The 2013 defeat condition and the 2015 survival condition are incomplete. The universe's region can be the whole history while genuine subsystems have proper regions, and the existence of subsystems does not make P1** true of the universe. The empirical question about subsystems is not the question on which the view turns.

The retreat to subsystems gives up the global claim, and even for subsystems (a) and (b) do not imply a proper union. The missing premise is a selection of which unities' presents count, not another bound on the extent of presents. Brading's principle is scale-free, so it cannot make that selection by itself. A presentist who wants the global restriction must add a privileged actuality, a co-presence relation, or a relative existence predicate. Each is a further structure with a cost; none is P1**.

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